Exercise Sheet 3
Propositional Logic – Syntax & Natural Deduction

Note that question 4 is marked as being assessed.

1. Let us write s for "I'm eating a steak"; f for "I'm eating fries"; c for "I'm eating chicken"; p for "I'm eating peas"; i for "I'm eating an ice cream". Turn the following sentences into propositional logic formulas:
   - Example: "If I eat a steak then I also eat fries and an ice cream" would be $s \to (f \wedge i)$
   - "If I eat fries then I don't eat an ice cream"
   - "If I eat a steak without fries then I eat it with peas"
   - "If I eat chicken with peas then I also eat some fries"
   - "I never eat both steak and chicken at the same time"
   - "It is not true that I never eat steak and fries at the same time"

2. Consider the following simplified version of the Sudoku puzzle. Consider a 2 by 2 matrix:

p	q
r	s

   It has 4 cells called p, q, r, and s. The goal is to fill each cell with either a 0 or a 1 such that
   - each row has exactly one 0 and one 1
   - each column has exactly one 0 and one 1

   Let the atomic proposition p_0 stand for "cell p is filled with 0"; p_1 stand for "cell p is filled with 1"; q_0 stand for "cell q is filled with 0"; q_1 stand for "cell q is filled with 1"; r_0 stand for "cell r is filled with 0"; r_1 stand for "cell r is filled with 1"; s_0 stand for "cell s is filled with 0"; s_1 stand for "cell s is filled with 1". Formalize the above rules as a propositional logic formula.

3. Provide a Natural Deduction proof of $A \to B \to A$

4. **assessed:** Provide a Natural Deduction proof of $(A \wedge B) \to \neg A \to C$

5. Provide a Natural Deduction proof of $(A \to B) \to (C \to \neg B) \to C \to \neg A$

6. We defined the syntax of propositional logic using the following grammar rule:

   $$P ::= a \mid P \wedge P \mid P \vee P \mid P \to P \mid \neg P$$

   Add if-and-only-if (iff) formulas of the form $P \leftrightarrow Q$ to this language. Informally, $P \leftrightarrow Q$ is true if both P implies Q, and Q implies P. What introduction and elimination rules can you add to the Natural Deduction proof system we have seen so far to allow reasoning about such formulas?